

# Rahul Shah

<https://github.com/rsha256/>

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## EDUCATION

### University of California, Berkeley

**GPA: 4.0**

M.S. Electrical Engineering & Computer Science. Thesis: *Optimizing ML Systems*

**Expected Graduation:** May 2025

B.A. Computer Science; B.A. Mathematics; B.A. Statistics

August 2020 - May 2024

Honors: Dean's List, Honors to Date, CS Honor Society (UPE) Officer, EECS Honors, Higher Honors in Mathematics

Activities & Societies: CSUA (CS Club Executive Officer), CS Mentors (CS 61A/B/70/88 Tutor), IEEE, ICPC.

**Relevant Coursework:** Data Structures & Algorithms<sup>†</sup>, Natural Language Processing<sup>†</sup>, Programming Languages & Compilers<sup>†</sup>, Machine Learning, Neural Nets<sup>†</sup>, Deep Reinforcement Learning<sup>†</sup>, Data Mining, Parallel Systems<sup>†</sup>, Computer Vision<sup>†</sup>, Human-Computer Interaction<sup>†</sup>, Computer Graphics<sup>†</sup>, Probability Theory<sup>†</sup>, Forecasting<sup>†</sup>, Optimization Methods<sup>†</sup>. <sup>†</sup>= Graduate

## EXPERIENCE

- **Schonfeld Strategic Advisors** (Quantitative Hedge Fund) New York, NY  
*Alpha Quantitative Researcher Intern* Apr 2023 - Aug 2023
  - Optimized >100 GB of alternative data from AWS S3 & EC2. Exploratory Data Analysis using **Databricks'** PySpark, for a statistically backtested scalable signal (currently in production) with Sharpe Ratio 1.8. Under Systematic Trading team (QIS).
  - Wrote an **Online** Algorithm to ingest live data & improve latency by 53% via Statistical Estimation for a HFT Stat Arb strategy. Measured effects of out-of-sample distributional shift via **divergence** measures, for a 12x improvement.
- **Arrowstreet Capital LP** | *Research Core* Boston, MA  
*Quantitative Developer Intern* Jun 2022 - Aug 2022
  - Fine-tuned a local **FinBERT** model from **Hugging Face** on over 100,000 vendor filings of various languages & correlated sentiments with word embeddings using **Cosine Similarity**, Jaccard Index, and Minkowski Distances.
  - Wrote distributed code with **Numba** and **Dask** to discover hidden signals for equity alpha research. Made robust improvements to the firm-wide single systematic process with equities toolings via **Bayesian Regressions** saving \$15 million annually.
- **Electrical Engineering & Computer Sciences** | **University of California, Berkeley** Berkeley, CA  
*Researcher;* *Teaching: Instructor, Head Teaching Assistant, Graduate Student Instructor.* Jan 2021 - Present
  - **Summer 2024 Instructor - CS 160 (Human-Computer Interaction):** Lectured Contextual Inquiry, creating user flows, wireframes, and building user interface mockups/prototypes). Advanced Javascript coding and frameworks.
  - **Lead TA - CS C100/200 (Data Science/Machine Learning), STAT 165/265 (Forecasting), CS 164 (Compilers):** Managed admin logistics for ~300 students via the EdStem Q&A platform, assignment extensions (automated using Gradescope API), Incomplete linkages + communications, held OH, taught graduate sections.
  - **Head Teaching Assistant - EECS 122 (Communication Networking/Internet):** Lead all administrative course logistics, **created** all existing course **content**, held office hours and led section on-campus.
  - **Berkeley Research - Artificial Intelligence:** Deep Learning & NLP: Trained an **LSTM** to predict student grades from attendance and analyzed **Time Series** to determine earnestness with new SOTA **metrics** (Paper accepted in ACM's 2023 conference: <https://arxiv.org/abs/2311.10746>). Also applied Deep RL techniques to 'solve' Minecraft. Peer reviewed abstract available: [cmt3.research.microsoft.com/api/DEEPRLFPP2022/Files/124](https://cmt3.research.microsoft.com/api/DEEPRLFPP2022/Files/124).

## PROJECTS

- **Live Streaming NeRF Nerfstudio Plugin:** Live re-rendering of shaders and textures in a game environment in real-time with minimal latency, with efficient C++ code, as well as adaptive noise reduction with anisotropic filters. Used a novel networking protocol to parse and transmit the live data from PyTorch, without loss or unmatched data pairs.
- **GPU-accelerated Factorization Machines:** Boosted recommender systems training speed by **21.4%** using a novel randomized algorithm that capitalized on data **sparsity** to parallelize Stochastic Gradient Descent. Achieved via custom CUDA structures for efficient **low-rank** polynomial kernel approximation. Profiled bottlenecks with VTune's Roofline.
- **Robust automated cable insertion system for online environments:** Applied CV and flood-fill algorithms for cable & channel identification using perception, segmentation, tracing & endpoint **detection**. Collaborated with Simeon Adebola (PhD candidate) and Professor Goldberg from AUTOLab and Berkeley AI Research (BAIR) Lab.

## AWARDS

- **Donald M. Rizzo Memorial Scholarship Recipient:** Awarded from international NPO: Lions Club International
- **Winner of the Congressional App Challenge:** Invited to Capitol Hill to demo mobile app to House of Representative members at #HouseOfCode, the most prestigious student STEM expo, at the U.S. Capitol Building
- **Capture The Flag Security Competitions:** Ranked **top 50** in all of {PicoCTF, HSCTF, BACTF, CSAW Red}

## SKILLS

- **Languages:** Python, Java, Go, C++, JavaScript/Node.js, Rust, Scala, C, TypeScript, Objective-C, C#, Shell (bash, zsh), OCaml, R.
- **Technologies:** AWS, Databricks, Heroku, GCP; Docker/Kubernetes; Parquet, MySQL/PostgreSQL, MongoDB; React, Vue.
- **AI/ML:** PyTorch, Keras, Tensorflow; LASSO, ANOVA, Kalman Filters; CNN, R-CNN, GNN, Transformer, BERT, SSM, RAG.